QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 1

RULE · Multi-step equations (ko'p qadamli tenglama)

Order: clear fractions → expand → collect the variable → combine like terms.

$$\frac{2}{3}(x+3) = x - 1 \implies x = 9$$

RULE · Sign-flip rule (ishora almashish qoidasi)

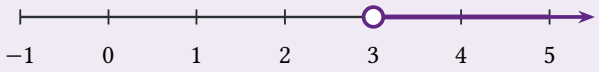
Operation	Sign flips?
Add or subtract	No
Multiply / divide by a positive	No
Multiply / divide by a negative	**Yes**

$$-3x + 5 \le 20 \implies x \ge -5$$

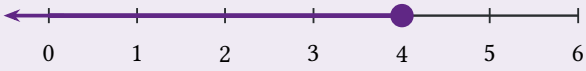
RULE · Graphing the solution

Sign	Circle	Meaning
$<$ or $>$	Open circle	Boundary excluded
$\le$ or $\ge$	Closed circle	Boundary included

$x > 3$ **(open)**



$x \le 4$ **(closed)**

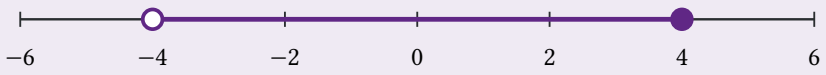


RULES SHEET – 2/2

RULE · Compound inequalities (qo'shma tengsizlik)

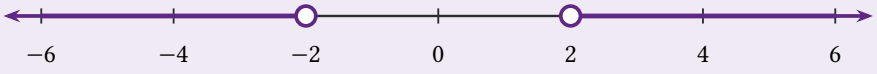
"**And**" – one operation, all three parts.

$-3 < x + 1 \leq 5 \implies -4 < x \leq 4$



"**Or**" – each part solved separately.

$2x + 1 < -3$ or $2x + 1 > 7 \implies x < -2$ or $x > 3$



Operation applies to all three parts of an "and" inequality – not just one.

RULE · Interval notation (oraliq belgilash)

Traditional textbooks: semicolon. SAT / English sources: comma.

Traditional: $(2;\ 5]$ SAT: $(2,\ 5]$